- **Scene:** Table with a hook, red cube, and blue cube
- **Start:** All objects (hook, red cube, blue cube) are in workspace
- **Goal:** Put the red cube where the blue cube is
- **Action Skeleton:** pick(blue_cube) → place(blue_cube) → pick(red_cube) → place(red_cube)

6. **Rearrangement Memory (Task 2):**
   - **Scene:** Table with a hook,red cube, and blue cube
   - **Start:** Hook and blue cube are in workspace, red cube is beyond workspace
   - **Goal:** Put the red cube where the blue cube is
   - **Action Skeleton:** pick(hook) → pull(red_cube, hook) → place(hook) → pick(blue_cube) → place(blue_cube) → pick(red_cube) → place(red_cube)

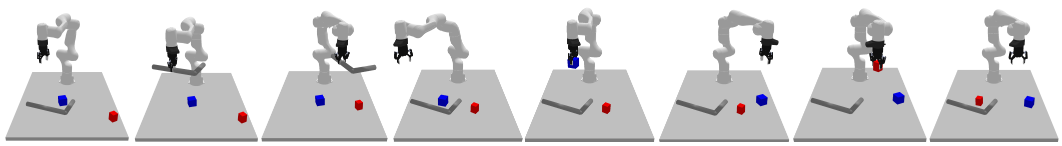

Figure 12: Rearrangement Memory Task 2

E.1 Skill Structure

We consider a finite set of parameterized skills in our skill library. The parameterization, data collection, and training method for each of the skills is described as follows:

1. `Pick`: Gripper picks up an object from the table and the parameters contain 4-DoF pose in the object's frame of reference (x, y, z, θ).
2. `Place`: Gripper places an object at the target location and parameters contain 4-DoF pose in the place target's frame of reference (x, y, z, θ). This skill requires specifying two set of parameters, the target pose and the target object (e.g. hook, table).
3. `Push`: Gripper uses the grasped object to push away another object. The skill is motivated from prior work [43, 1] where a hook object is used to `Push` blocks. The parameters of this skill are (x, y, r, θ) such that the hook is placed at the (x, y) position on the table and pushed by a distance r in the radial direction θ w.r.t. the origin of the manipulator.
4. `Pull`: Gripper uses the grasped object to pull another object inwards. The skill is also motivated from prior work [43, 1] where a hook object is used to `Pull` blocks. The parameters of this skill are (x, y, r, θ) such that the hook is placed at the (x, y) position on the table and pulled by a distance r in the radial direction θ w.r.t. the origin of the manipulator.

E.2 State Space of the Unified Skill Transition Model

CDGS assumes access to 6D object poses. In practice, we construct the system state as a concatenated vector of poses of objects present in the scenario. We use a fixed object order ([robot, rack, hook, cube1, cube2, . . .]), passing zero-vectors for absent objects, consistent across all baselines for the experiment.